# ARF-Plus: Controlling Perceptual Factors in Artistic Radiance Fields for 3D Scene Stylization

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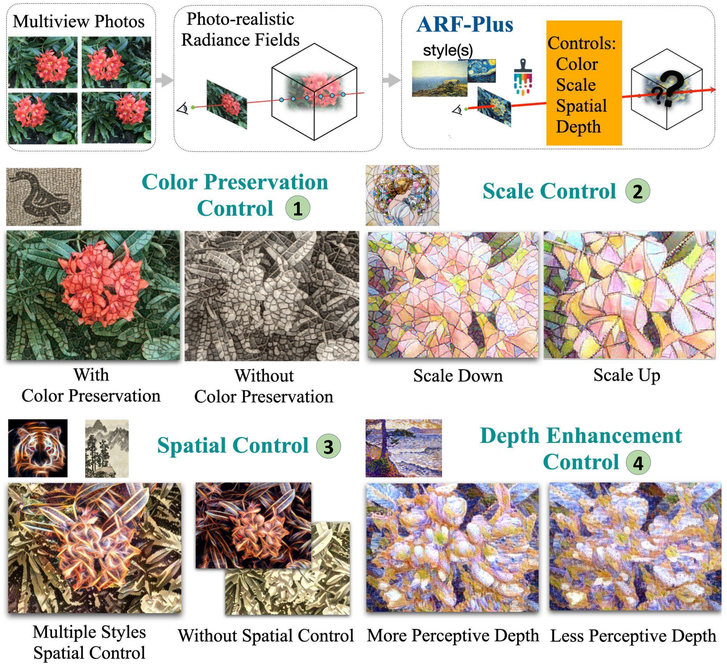
## Overview

**Research Gap and Motivation**: We identify a research gap on radiance fields style transfer, emphasizing the importance of diverse perceptual controls in 3D stylization to meet varied user requirements across multiple application scenarios.

**Proposed Framework:** We introduce ARF-Plus, an innovative framework enabling simultaneous manipulation of multiple perceptual factors in 3D stylization, , pioneering both practical implementation and innovation in radiance fields.

**Evaluation and Results**: We conduct extensive qualitative and quantitative evaluations, showcasing the effectiveness and versatility of the ARF-Plus framework.

## Method

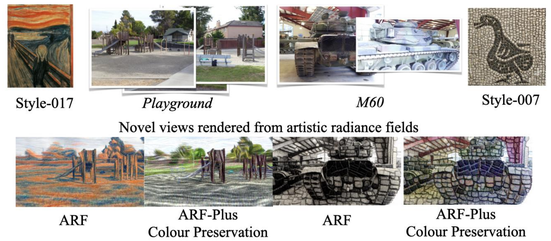



Fine-tune photo-realistic radiance fields using given style images, framing style transfer as an **optimization problem**. Utilize *tailored loss functions* and *gradient update strategies* to implement selected perceptual controls.

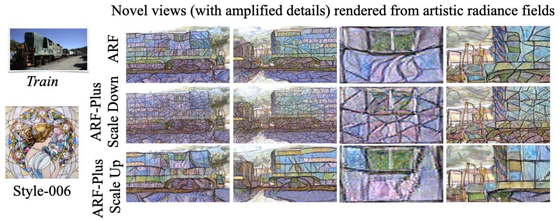
1. luminance channel -> $L_{style}$
2. the receptive field + the size of the style image -> $L_{style}$
3. cached-gradients map + spatial mask
4. pre-trained depth estimation network -> $L_{depth}$

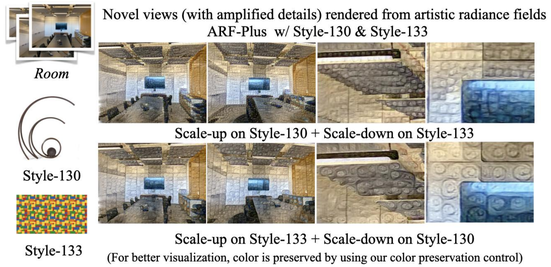
## Results

### Color Preservation Control

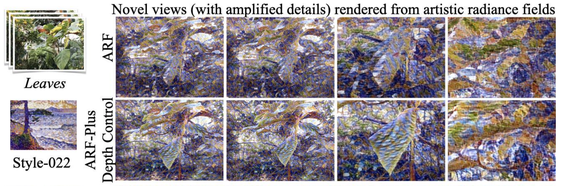



### Single / Multiple Style(s) Scale Control



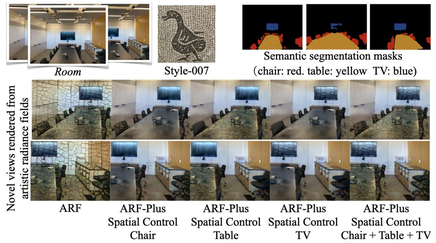



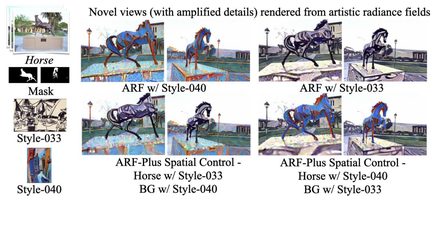

### Depth Enhancement Control



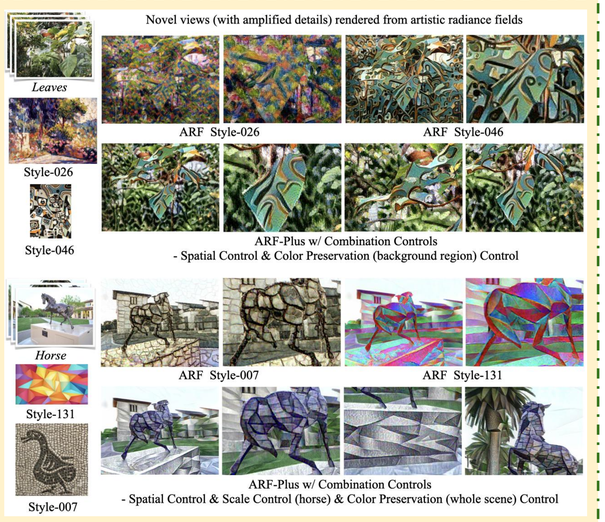

### Single / Multiple Style(s) Spatial Control

| Style | Scene | ArtFID ↓ ARF in Mask | ArtFID ↓ ARF-Plus w/ Spatial Control in Mask | ContentDist ↓ ARF in Mask | ContentDist ↓ ARF-Plus w/ Spatial Control in Mask | StyleFID ↓ ARF in Mask | StyleFID ↓ ARF-Plus w/ Spatial Control in Mask |
|---|---|---|---|---|---|---|---|
| 007 | *Room*-Chair | 41.4614 | **39.3775** | 0.0435 | **0.0434** | 38.7340 | **36.7384** |
| 007 | *Room*-Table | 43.2753 | **40.8616** | **0.0473** | 0.0489 | 40.3191 | **37.9581** |
| 007 | *Room*-TV | 41.2685 | **38.4802** | 0.0195 | **0.0193** | 39.4799 | **36.7500** |
| 007 | *Room*-All | 41.0287 | **37.9573** | **0.0950** | 0.0983 | 36.4707 | **33.5591** |




### Combination of Controls

Enabling custom stylization and flexible blending of different styles to create unique and striking effects for 3D scenes

**Greater Flexibility + Endless Possibility**